

Esp32Tap Breadboard — Module Test Checklist

Print copy · Current-limited bench bring-up · Preflight plus Steps 1–10

Do not connect the treadmill until Step 10. Turn all power off before moving a wire or removable jumper.

Preflight — unpowered audit

Required state: USB disconnected · bench supply disconnected · treadmill/RJ45 cables disconnected · **STANDALONE POWER removed** · **COIL POWER removed**.

Rails and polarity

Meter check	Expected	Measured
+8V_RAW → GND	Not shorted	
VIN → GND	Not shorted	
TSR_3V3 → GND	Not shorted	
LOGIC_3V3 → GND	Not shorted	
+5V_RLY → GND	Not shorted	
Unlike positive rails	Not joined	

☐ 1N5822 stripe/cathode faces VIN

☐ Electrolytic negative stripes face GND

☐ TPS3700/TPS709 adapters were continuity-mapped

Relay identification

Unpowered check	Expected	Measured
K1 pins 1–16	≈50 Ω coil	
K1 pins 4–6	Continuity	
K1 pins 8–6	Open	
K1 pins 13–11	Continuity	
K1 pins 9–11	Open	

The relay datasheet drawing is a bottom view. Trust the meter-verified physical mapping.

STOP: Do not apply power if any positive rail remains near zero ohms to GND, if two unlike positive rails are joined, or if a polarized part is reversed.

Permanent source rules

- ☐ STANDALONE POWER is removed whenever USB is connected.
- ☐ COIL POWER remains removed until Step 6 passes.
- ☐ Bench measurements use the common GND rail as reference.
- ☐ If the supply reaches current limit, power is removed immediately.
- ☐ Any unexpected reset during a powered bench or treadmill test is an immediate STOP.

Operator:

Date:

Preflight — RJ45 paths and final visual comparison

Pins 1/7 share GND and pins 2/8 share +8V_RAW only after their four final common links are installed. If isolated construction evidence was not recorded, power off and temporarily lift those four common links to perform the independent tests below.

Independent conductor	Pass requirement	Measured / evidence
CONSOLE.1 → MOTOR.1	<2 Ω; all other isolated contacts open	
CONSOLE.7 → MOTOR.7	<2 Ω; all other isolated contacts open	
CONSOLE.2 → MOTOR.2	<2 Ω; all other isolated contacts open	
CONSOLE.8 → MOTOR.8	<2 Ω; all other isolated contacts open	
CONSOLE.3 → MOTOR.3	<2 Ω; no cross-short	
CONSOLE.4 → MOTOR.4	<2 Ω; no cross-short	
CONSOLE.5 → MOTOR.5	<2 Ω; no cross-short	
CONSOLE.6 → MOTOR.6 through K1 NC	<2 Ω; K1 NO/TX path open	

- ☐ Every CONSOLE pin number was continuity-mapped to its breakout terminal
- ☐ Every MOTOR pin number was continuity-mapped to its breakout terminal
- ☐ No unintended shorts exist between different RJ45 nets
- ☐ Four final common links were restored and verified: CONSOLE/MOTOR pins 1/7 → GND; pins 2/8 → +8V_RAW

Final unpowered visual comparison

- ☐ Full assembly compared against clustered layout and point-to-point list
- ☐ DIP notches, adapter pin-1 marks, transistor leads, diodes, LEDs, and electrolytics checked
- ☐ No loose strands, adjacent-row bridges, or parts sharing an unintended five-hole strip
- ☐ Overhead photograph/evidence reference recorded below

Photo or evidence reference:

STOP: Any conductor exceeds 2 Ω, any unrelated contact is shorted, a mapping is uncertain, or final visual comparison disagrees with the wiring list.

Step 1 — Raw-input protection

USB: disconnected · **STANDALONE POWER:** removed · **COIL POWER:** removed · bench supply to +8V_RAW/GND at **8.00 V**, **250 mA** limit.

+8V_RAW → RXEF075 → FUSED_8V → 1N5822 → VIN

Point	Pass range	Measured
+8V_RAW	About 8.00 V	
FUSED_8V, after RXEF075	Close to 8.00 V	
VIN, after 1N5822	7.20–7.90 V	
Total bench-supply current	Below 50 mA	

☐ Protection chain passed

STOP: Current rises toward 250 mA; VIN is zero/negative; or diode, polyfuse, TVS, or wiring becomes warm.

Step 2 — Standalone 3.3 V regulator

Keep the Step 1 configuration. **Do not install STANDALONE POWER.**

VIN → TSR 1-2433E → TSR_3V3 [open jumper] LOGIC_3V3

Point	Pass range	Measured
TSR pin 1 input	Same as VIN	
TSR pin 3 / TSR_3V3	3.20–3.40 V	
LOGIC_3V3	Approximately 0 V	

☐ Input capacitor + is VIN; – is GND

☐ Output capacitor + is TSR_3V3; – is GND

☐ TSR module passed

STOP: TSR_3V3 is outside 3.20–3.40 V, LOGIC_3V3 is unexpectedly powered, polarity is wrong, or current/temperature rises.

Step 3 — USB-powered logic rail

Disconnect bench supply. Connect USB only to the DevKit port labeled **UART**. STANDALONE POWER and COIL POWER remain removed.

Point	Pass range/state	Measured
DevKit 3V3	3.20–3.40 V	
LOGIC_3V3 rail	3.20–3.40 V	
TSR_3V3	Approximately 0 V	
RELAY_CMD / TX_ENABLE	Both low	
RELAY_GATE / TX_GATE	Both low	
+5V_RLY	Below 0.25 V	

☐ Red LOGIC_3V3 power LED illuminates

☐ USB logic stage passed

STOP: LOGIC_3V3 is outside 3.20–3.40 V, TSR_3V3 is energized through the open link, any control/gate is unexpectedly high, or +5V_RLY exceeds 0.25 V.

Step 4 — TPS3700 voltage window

USB remains connected. Bench supply returns to +8V_RAW/GND. STANDALONE POWER and COIL POWER remain removed.

At normal VIN	Expected	Measured
Protected VIN	Record actual value	
UV_SENSE	$VIN \div 16$	
OV_SENSE	$VIN \div 26.5$	
TREAD_OK near VIN=8 V	High, normally ≈ 3.0 V; >2.5 V	
GPIO6 side of 4.7 k Ω	Same logical state	

Sweep using measured VIN	Required TREAD_OK	Observed transition
Below UV window	Low, <0.25 V	
VIN rising through 6.25–6.55 V	Transitions high	
VIN around 8.0 V	High, >2.5 V	
VIN above 10.90 V but no more than 11.1 V	Low, <0.25 V	
Raise VIN above OV, then lower through 10.30–10.90 V	Transitions high on falling boundary	

STOP: Do not exceed protected VIN ≈ 11.1 V. Stop if the TVS draws appreciable current or becomes warm.

Step 5 — AHC08 hardware permission gates

RELAY_GATE = RELAY_CMD AND TREAD_OK
TX_GATE = TX_ENABLE AND TREAD_OK

The current no-control diagnostic firmware cannot raise RELAY_CMD or TX_ENABLE. Before manually driving either input, disconnect its DevKit jumper: GPIO21 for RELAY_CMD or GPIO15 for TX_ENABLE. Never drive against a connected GPIO output.

Use a temporary LOGIC_3V3 jumper on the disconnected command input. Observe command, TREAD_OK, and gate output with the logic analyzer.

Test	Command	TREAD_OK	Expected gate	Observed
Inhibit	0	0	0	
Hardware override	1	0	0	
No command	0	1	0	
Permitted command	1	1	1	

☐ RELAY_GATE truth table passed

☐ TX_GATE truth table passed

☐ Temporary drives removed; keep GPIO21 disconnected only if proceeding directly to the unloaded Step 6 test

STOP: Either output becomes high unless both its command and TREAD_OK are high, or a manual drive is applied while the DevKit GPIO remains connected.

Step 6 — Unloaded TPS709 relay supply

VIN must be inside the valid TREAD_OK window. **COIL POWER remains removed.** Command using either the qualified exerciser or a supervised temporary LOGIC_3V3 drive with the GPIO21 jumper disconnected.

State / point	Pass range	Measured
Command off: RELAY_GATE	Below 0.25 V	
Command off: +5V_RLY	Below 0.25 V	
Command on: RELAY_GATE / TPS709 EN	Approximately 3.3 V	
Command on: +5V_RLY	4.75–5.25 V	
Command on: BC337 base	Approximately 0.65–0.90 V	
Command off again: +5V_RLY	Discharges below 0.25 V	

☐ Command is off and temporary drive is removed

☐ With power off, GPIO21 jumper is restored unless qualified exerciser firmware will be used

☐ Unloaded relay supply passed

STOP: Disabled +5V_RLY exceeds 0.25 V, enabled +5V_RLY is outside 4.75–5.25 V, the rail fails to discharge, or COIL POWER is installed during this step.

Step 7 — Relay coil, transistor, and contacts

Proceed only after Step 6 passes and identified bounded relay-exerciser firmware is installed. Do not energize the coil with a latched manual jumper. The current no-control diagnostic image cannot perform this step.

Bench-exerciser state: Pi UART USB connected · STANDALONE POWER removed · bench supply at +8V_RAW/GND · COIL POWER installed only after the power-off sequence below · command and observations over Pi UART.

Relay-exerciser build ID:

- 1. Turn off USB and bench power.
- 2. Confirm +5V_RLY is below 0.25 V.
- 3. Install COIL POWER.
- 4. Reconnect Pi UART USB; verify the exact relay-exerciser build ID and that the relay command is off.
- 5. Set +8V_RAW to 8.00 V with a maximum 500 mA current limit.
- 6. Issue one bounded relay-on command, then verify it removes itself.

Energized measurement	Pass range	Measured
Voltage across K1 pins 1–16	At least 4.50 V	
Coil current	90–110 mA	
BC337 collector-to-emitter	No more than 0.30 V	
+5V_RLY under load	Record; must support coil requirement	

Relay state	K1 4–6	K1 8–6	GPIO4, GPIO5	Observed
Off / bypass	Closed	Open	0, 1	
On / ESP drive	Open	Closed	1, 0	

STOP: Feedback pair **00** or **11** is always a fault.

Step 7 continued — fail-release and thermal hold

Fail-release tests — logic analyzer timing

For each row, begin energized, remove only the named permission/source, and measure time until NC bypass is restored and feedback settles to (0,1). Restore power/state only after the relay is off.

Removed while energized	Pass requirement	Measured
Pi UART USB / logic power	NC restored ≤ 100 ms; GPIO4/5 $\rightarrow$ (0,1)	
GPIO21 / RELAY_CMD connection	NC restored ≤ 100 ms; GPIO4/5 $\rightarrow$ (0,1)	
TREAD_OK permission	NC restored ≤ 100 ms; GPIO4/5 $\rightarrow$ (0,1)	
VIN / bench supply	NC restored ≤ 100 ms; GPIO4/5 $\rightarrow$ (0,1)	

Five-minute thermal hold — only after short pulses pass

Part	Ambient	After 5 minutes	Pass limit
TPS709 case			$\leq 45^{\circ}\text{C}$ and $\leq 10^{\circ}\text{C}$ rise
BC337 case			$\leq 45^{\circ}\text{C}$ and $\leq 10^{\circ}\text{C}$ rise

STOP: Supply limits, voltage/current limits, contact truth, or temperature limits fail; any component heats rapidly; any reset occurs; or relay does not return immediately to bypass when command is removed.

☐ Relay stage passed

Step 8 — AHC126 UART isolation

Treadmill remains disconnected. Use the logic analyzer on TX_GATE, AHC126 pins 2/3, K1 pin 8, K1 pin 6, and console pin 6 as channels allow.

Condition	Expected	Observed
TX_GATE low	AHC126 pin 3 is high impedance	
TX_GATE high	AHC126 pin 3 follows pin 2	
Relay off	Console pin 6 passes to motor pin 6	
Relay off	TX_DRV/K1 pin 8 isolated from motor pin 6	
Relay on	Motor pin 6 connects to buffered ESP TX	
Relay on	Console pin 6 isolated from motor pin 6	

Firmware gate: Dynamic transmit testing requires identified UART/relay exerciser firmware. The current safe diagnostic image deliberately does not transmit or energize the relay.

- ☐ Static isolation/continuity passed
- ☐ Dynamic test deferred until qualified exerciser exists, or qualified build ID recorded:

Build ID:

STOP: TX appears at AHC126 pin 3 while TX_GATE is low, relay contact isolation disagrees with the table, UART idle is missing, or required exerciser identity is absent.

Step 9 — Standalone-power boot

1. Flash and record identified standalone observer firmware while USB diagnostic power is still available.
2. Turn off all power and physically disconnect USB.
3. Install STANDALONE POWER. Install COIL POWER only if Step 7 passed. Relay and TX must remain disabled.
4. Apply 8.00 V to +8V_RAW with an initial 500 mA limit.

Standalone observer build ID:

Manifest path/SHA proving relay and TX outputs disabled:

☐ Build manifest was inspected and explicitly proves relay and TX outputs are disabled

Point	Pass range/state	Measured
VIN	7.20–7.90 V	
TSR_3V3	3.20–3.40 V	
LOGIC_3V3	3.20–3.40 V	
DevKit	Boots; red power LED on	
Relay / GPIO4,GPIO5	Off/bypass; feedback (0,1)	
RELAY_CMD / TX_ENABLE	Both disabled	
Wi-Fi observation API	Reachable; bounded event log returned	
GPIO6 / TREAD_OK observation	Matches meter state	
GPIO16 UART idle, before UART configuration	Record measured logic level	
GPIO18 UART idle, before UART configuration	Record measured logic level	

Before reconnecting USB: turn off bench power and remove STANDALONE POWER.

☐ Standalone boot passed

STOP: Observer identity/manifest proof is absent, Wi-Fi/event-log observation fails, either relay/TX output enables, feedback is not (0,1), UART idle is missing, any reset occurs, or any rail is outside its limit.

Step 10 — First treadmill connection: bypass only

This is not a control test. Do not proceed unless Steps 1–9 pass and identified observer firmware is installed.

Required state before attachment

- ☐ Treadmill power is off
- ☐ USB is physically disconnected
- ☐ STANDALONE POWER is installed
- ☐ COIL POWER is installed only because the complete Step 7 bench relay test passed
- ☐ Qualified observer/no-control firmware build ID is recorded
- ☐ Relay is commanded off and NC bypass continuity is proven
- ☐ ESP transmission is disabled
- ☐ Console and motor RJ45 identities/pins are meter-verified

Observer build ID:

Powered bypass observations

Temporary dual-pin current harness — treadmill power off

- ☐ Remove both direct CONSOLE +8 V links for pins 2 and 8
- ☐ Join CONSOLE pins 2 and 8 only at the fused DMM current input
- ☐ Split the DMM current output back to two independent breadboard +8 V paths
- ☐ Meter leads are in COM and the fused current jack; current range selected
- ☐ Harness polarity and absence of a direct bypass around the meter are verified

Temporary-harness measurement	Pass limit	Measured
Total +8 V current using temporary fused-DMM harness	No more than 500 mA	

After measuring current: treadmill power off → verify zero current → remove temporary ammeter harness → restore both direct CONSOLE pin 2/8 paths → move meter lead out of current jack → then power for voltage-drop measurements.

- ☐ Both direct +8 V paths restored and individually continuity-verified

After direct paths are restored	Pass limit	Measured
Console-to-motor supply drop	No more than 50 mV	
Motor-to-console ground-return drop	No more than 50 mV	
Relay state	Off / NC bypass	
Console pin 6 to motor pin 6	Continuous in bypass	
UART observation	Receive/observe only; no transmit	

Step 10 continued — thermal audit and final gate

15-minute thermal observation — inspect every terminal

RJ45 pin	CONSOLE temperature	MOTOR temperature	Pass limit at each
1 — GND			≤40°C and ≤10°C rise
2 — +8 V			≤40°C and ≤10°C rise
3			≤40°C and ≤10°C rise
4			≤40°C and ≤10°C rise
5			≤40°C and ≤10°C rise
6 — switched data			≤40°C and ≤10°C rise
7 — GND			≤40°C and ≤10°C rise
8 — +8 V			≤40°C and ≤10°C rise

Additional location—all endpoints checked	Ambient	Highest after 15 min	Pass limit
All +8 V jumper endpoints/breadboard contacts			≤40°C and ≤10°C rise
All GND jumper endpoints/breadboard contacts			≤40°C and ≤10°C rise

Hottest exact terminal/jumper endpoint and value:

STOP: Any current, voltage-drop, temperature, bypass-continuity, firmware-identity check fails, or any reset occurs. No relay transfer is permitted during this phase.

☐ Bypass-only treadmill test passed

Bench Record Summary

A failed step stops the chain. Diagnose that module before continuing downstream.

Stage	Pass	Date/time	Notes / failure reference
Preflight — unpowered audit	☐		
1 — Raw-input protection	☐		
2 — Standalone 3.3 V regulator	☐		
3 — USB-powered logic rail	☐		
4 — TPS3700 voltage window	☐		
5 — AHC08 permission gates	☐		
6 — Unloaded TPS709 relay supply	☐		
7 — Relay coil/transistor/contacts	☐		
8 — AHC126 UART isolation	☐		
9 — Standalone-power boot	☐		
10 — Treadmill bypass only	☐		

Equipment

Bench supply:

Multimeter:

Logic analyzer:

Ambient temperature:

DevKit serial:

Firmware build ID:

Notes

[illegible]

Source handoff reminder: USB connected ⇒ STANDALONE POWER removed. STANDALONE POWER installed ⇒ USB physically disconnected.

Final reviewer:

Date: